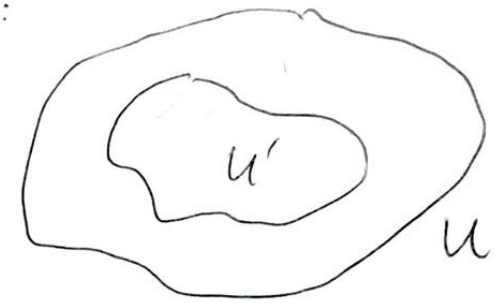
<u>Schauder estimate 2</u> Let $U' \subset\subset U \subset \mathbb{R}^n$ be open ($\overline{U'} \cap U$ compact)

$k \in \mathbb{N}$, $\alpha \in (0,1)$, $\Lambda > 0$

L 2nd order elliptic operator:



$$Lf = \sum_{i,j} a_{ij} \partial_i \partial_j f + \sum_i b_i \partial_i f + cf$$

$a_{ij}, b_i, c : U \to \mathbb{R}$

$a_{ij} = a_{ji}$, $\quad \Lambda^{-1} \delta_{ij} < a_{ij} < \Lambda \delta_{ij}$ (ellipticity)

$\|a_{ij}\|_{C^{k,\alpha}(U)}, \|b_i\|_{C^{k,\alpha}(U)}, \|c\|_{C^{k,\alpha}(U)} < \Lambda$

Then $\|f\|_{C^{k+2,\alpha}(U')} \leq C(\|Lf\|_{C^{k,\alpha}(U)} + \|f\|_{C^0(U)})$

C depends on $U', U, k, \alpha, \Lambda$

Regularity statement: if $f \in C^2$, and $Lf \in C^{k,\alpha}$
$\Rightarrow f \in C^{k+2,\alpha}$

Schauder estimate on compact manifold

Let M be a compact mfld.

L = 2nd order elliptic operator i.e.

in charts $Lf = \sum a_{ij}\partial_i\partial_j f + \sum b_i \partial_i f + cf$

with same assumptions on a_{ij}, b_i, c as above

Then $L : C^{k+2,\alpha}(M) \longrightarrow C^{k,\alpha}(M)$

is a bounded linear map

and $\|f\|_{C^{k+2,\alpha}(M)} \leq C\left(\|Lf\|_{C^{k,\alpha}(M)} + \|f\|_{C^0(M)}\right)$

<u>Proof</u> Apply local Schauder estimate in charts: cover $M = \bigcup_{i=1}^{N} U_i$

and let $U_i' \subset\subset U_i$ s.t. U_i' still cover M

$$\Rightarrow \|f\|_{C^{k+2,\alpha}(U_i')} \leq C\left(\|Lf\|_{C^{k,\alpha}(U_i)} + \|f\|_{C^0(U_i)}\right)$$

$$\leq C\left(\|Lf\|_{C^{k,\alpha}(M)} + \|f\|_{C^0(M)}\right)$$

$$\Rightarrow \|f\|_{C^{k+2,\alpha}(M)} \leq C\left(\|Lf\|_{C^{k,\alpha}(M)} + \|f\|_{C^0(M)}\right)$$

$\square$

Laplacian on a Riemannian manifold

On (M,g) define $\Delta f = \sum_{j,k} g^{jk} \nabla_j \nabla_k f$

(In Kähler case we usually define
$$\Delta f = \sum_{j,k} g^{j\bar{k}} \nabla_j \nabla_{\bar{k}} f$$
which is half of the Riemannian version)

In local coordinates
$$g^{jk} \nabla_j \nabla_k f = g^{jk} \nabla_j \partial_k f$$
$$= g^{jk} \left(\partial_j \partial_k f - \Gamma^p_{jk} \partial_p f \right)$$

Schauder estimates apply as long as we have bounds of form
$$\Lambda^{-1} \delta_{ij} < g_{ij} < \Lambda \delta_{ij}, \quad \| g_{ij} \|_{C^{k,\alpha}}, \| \Gamma^p_{jk} \|_{C^{k,\alpha}} < \Lambda$$

<u>Theorem</u> Let (M,g) be compact,
$$\Delta : C^{k+2,\alpha}(M) \longrightarrow C^{k,\alpha}(M)$$
Then $\ker \Delta =$ constants
$$\operatorname{Im} \Delta = \left\{ f \in C^{k,\alpha}(M) \;\middle|\; \int_M f \, dV_g = 0 \right\}$$

Proof sketch

1. for f_1, f_2

$$\int_M f_1 \Delta f_2 \, dV = - \int_M g(\nabla f_1, \nabla f_2) \, dV$$

- Apply to $f_1 = f_2 = f$ s.t. $\Delta f = 0$:

$$0 = \int_M f \Delta f \, dV = - \int_M g(\nabla f, \nabla f) \, dV$$

$$\Rightarrow \nabla f = 0 \quad \text{i.e. } f = \text{const.}$$

- Apply to $f_1 = 1$, $f_2 = f$:

$$\int_M \Delta f \, dV = 0$$

$$\Rightarrow \operatorname{Im} \Delta \subset \left\{ f \;\middle|\; \int_M f \, dV = 0 \right\}$$

2. To prove $\operatorname{Im} \Delta = \left\{ f \;\middle|\; \int_M f \, dV = 0 \right\}$

Can use Schauder est. to show that $\operatorname{Im} \Delta \subset C^{k,\alpha}(M)$ is closed.

Then argue by duality to show that $\operatorname{Im} \Delta = \left\{ f \;\middle|\; \int_M f \, dV = 0 \right\}$

$\square$

Yau's theorem

(M, g) compact Kähler, $F \in C^\infty(M)$

$$\left.\begin{aligned} \det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f\right) &= e^{tF+c} \det(g_{j\bar{k}}) \\ g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f &> 0 \\ c &\in \mathbb{R} \end{aligned}\right\} \quad (*_t)$$

<u>Openness</u> If $(*_{t_0})$ admits a smooth solution (f_{t_0}, c_{t_0}) then $\exists\, \varepsilon > 0$ s.t. $(*_t)$ admits a smooth soln. for $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$.

<u>Proof</u> Consider the operator

$$F : \mathbb{R} \times \mathbb{R} \times C^{3,\alpha}(M) \longrightarrow C^{1,\alpha}(M)$$

$$(t, \lambda, \varphi) \longmapsto \log \frac{\det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}}(f_{t_0} + \varphi)\right)}{\det(g_{j\bar{k}})} - tF - (c_{t_0} + \lambda)$$

$$F(t_0, 0, 0) = 0$$

Linearization of F at $(t_0, 0, 0)$ with respect to λ, φ variables :

$$L(\lambda,\varphi) := DF_{(t_0,0,0)}(\lambda,\varphi) = \tilde{g}^{j\bar{k}}\partial_j\partial_{\bar{k}}\varphi - \lambda$$

$$\tilde{g}_{j\bar{k}} = g_{j\bar{k}} + \partial_j\partial_{\bar{k}} f_{t_0}$$

$$L : \mathbb{R} \times C^{3,\alpha}(M) \longrightarrow C^{1,\alpha}(M)$$
$$(\lambda,\varphi) \longmapsto \Delta_{\tilde{g}}\varphi - \lambda$$

is surjective.

Implicit function theorem

$\Longrightarrow$ $\exists\, \varepsilon > 0$ s.t. if $t \in (t_0-\varepsilon, t_0+\varepsilon)$

then there is (φ_t, λ_t) solving

$F(t, \lambda_t, \varphi_t) = 0$

Then $(f_{t_0} + \varphi_t,\ c_{t_0} + \lambda_t)$ solves $(*_t)$